

Q1. What are the 3 main connectivity modes in Power BI?

Power BI mainly provides three data connectivity modes:

1. Import Mode

Data is imported and stored in the Power BI model.

Advantages:

- Fast report performance.
- Supports extensive data modeling and DAX.
- Data can be refreshed on a schedule.

Example: Importing an Excel file into Power BI.

2. DirectQuery Mode

Data remains in the original database, and Power BI sends queries to the source when users interact with the report.

Advantages:

- Useful for very large datasets.
- Data can be more up-to-date because it is queried from the source.
- Reduces the amount of data stored in the Power BI model.

Example: Connecting Power BI directly to a large SQL Server database.

3. Live Connection

Power BI connects directly to an existing semantic model or Analysis Services model instead of importing the underlying data into Power BI Desktop.

Example: Connecting to an existing Power BI semantic model or SQL Server Analysis Services model.

Q2. What are the two gateways in Power BI?

The two types of Power BI gateways are:

1. On-premises Data Gateway

It allows Power BI Service to securely access on-premises data sources, such as SQL Server databases located inside an organization's network.

It can be used by multiple users and is suitable for organizational/enterprise scenarios.

2. Personal Mode Gateway

The On-premises data gateway (personal mode) is designed for an individual user. It is mainly used for personal data refresh scenarios.

Q3. How does a Gateway work? Explain with diagram.

A gateway acts as a bridge between Power BI Service in the cloud and an on-premises data source.

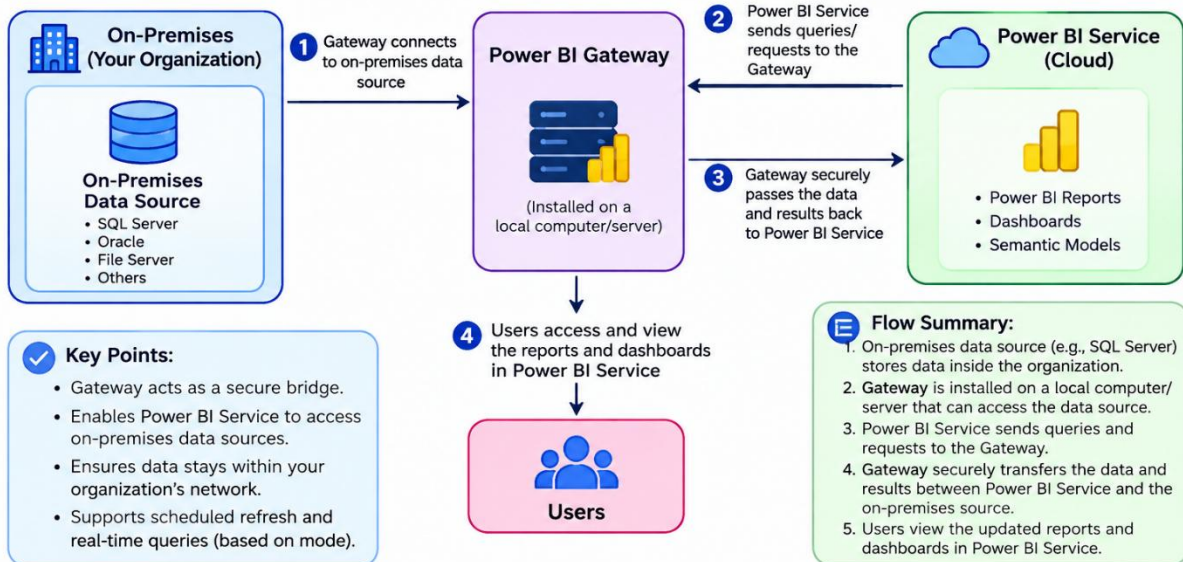
For example, suppose a company has a SQL Server database inside its office network. Power BI Service cannot directly access that private database. The gateway provides the connection.

Diagram:

How Power BI Gateway Works?



A Gateway acts as a bridge between Power BI Service (in the cloud) and on-premises data sources (inside an organization's network). It securely transfers data and queries between the cloud and on-premises systems.



How it works:

1. Data is stored in an on-premises database.
2. The gateway is installed on a computer/server that can access that database.
3. Power BI Service communicates with the gateway.
4. The gateway securely passes the required data/query information between Power BI Service and the on-premises source.
5. Power BI Service updates the semantic model or report according to the configured refresh/connectivity method.
6. Users view the updated report through Power BI Service.

Important: The gateway does not normally act as a permanent cloud storage location for your database. Its main role is to provide secure access between Power BI Service and supported on-premises sources.

Q4. What is the difference between Cloud-based Data and On-premises Data?

Cloud-based Data

Cloud-based data is stored on cloud servers and accessed through the internet.

Examples:

- Azure SQL Database
- Salesforce
- SharePoint Online
- Google BigQuery

Features:

- Accessible through the internet.
- Managed by the cloud provider or service.
- Usually doesn't require an on-premises gateway for Power BI to access the cloud source.

On-premises Data

On-premises data is stored on servers or computers within an organization's own infrastructure.

Examples:

- SQL Server installed in a company office.
- Local file server.
- On-premises Oracle database.

Features:

- Stored within the organization's network.
- Access may be restricted by firewall/network security.
- A gateway is generally required when Power BI Service needs to access the source.

Q5. What are the steps to use a Gateway?

Step 1: Download the Gateway

Go to Microsoft's Power BI Gateway download page and download the On-premises data gateway.

Step 2: Install the Gateway

Install it on a computer/server that can access your on-premises data source.

For example:

SQL Server



Gateway computer

Step 3: Sign in

Open the gateway application and sign in using your Power BI account.

Step 4: Register the Gateway

Choose the option to register a new gateway or add the computer to an existing gateway installation.

Step 5: Add the Data Source

In Power BI Service, go to the gateway management area and add your data source.

Provide details such as:

- Data source type
- Server name
- Database name
- Authentication method
- Credentials

Step 6: Connect the Semantic Model

Configure the Power BI semantic model to use the gateway/data source.

Step 7: Configure Refresh

Set up scheduled refresh if required.

For example:

On-premises SQL Server



Gateway



Power BI Service



Semantic Model



Report/Dashboard